Matej Ciglenečki

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CTO with over 4 years of experience, building a distributed machine learning system serving tens of thousands of active users. I'm a strong communicator who can present findings clearly and understandably. Throughout my academic and industry experience, I've developed excellent research and problem-solving capabilities accompanied by a strong sense of ownership. My research interests include generative modeling, vision-language models, video understanding, and NeRFs, with openness to related areas.

Education

M.Sc Data Science – Faculty of Electrical Engineering and Computing, Zagreb, Croatia — Oct 2021 – July 2023
master's thesis: **Generative Novel View Synthesis Using Neural Radiance Fields (NeRF)**
relevant courses: Advanced Algorithms and Data Structures, Machine Learning, Multivariate Data Analysis, Deep learning, Parallel Programming, Signal processing, Natural Language Processing, Numerical Mathematics, Deep Generative Modeling

B.Sc Computer Science – Faculty of Electrical Engineering and Computing, Zagreb, Croatia — Oct 2017 – July 2021
bachelor's thesis: **Facial Expression Recognition Using Convolutional Neural Networks**
relevant courses: Algorithms and Data Structures, OOP, Design Patterns, Databases, Probability and Statistics, Artificial intelligence

Experience

TensorPix – CTO — Dec 2024 – present
- Built and trained a model that predicts an image's degradation embedding (blur, compression, noise…) (**PyTorch**).
- Built a high-quality image filtering pipeline to filter training data, improving the super resolution model quality metrics.
- Developed a video inference engine for efficient video processing (**PyTorch, Python, ONNX, TensorRT**).
- Researched and benchmarked and torch.compile for the purpose of inference cold start with static and dynamic input shapes.
- Enhanced generative modeling benchmarking and evaluation suite (**PyTorch, Python**).
- Managed 3 engineers. Heavily influencing the company's strategy. Pitched to investors during Series A fundraising.

TensorPix – Machine Learning Engineer — Nov 2023 – Dec 2024
- Researched, implemented, and trained models for super resolution and low-light video enhancement (**PyTorch, Python**).
- Implemented a task queue and distributed workers for ML inference used in production (**Redis, Python, Django**).
- Implemented a custom cloud-agnostic instance auto scaler (**Redis, Python**).

Microblink – Machine Learning Engineer Intern — Feb 2023 – Sep 2023
- Researched and implemented NeRF to support the generation of segmentation masks via transfer learning. (**PyTorch**)
- Processed point clouds and images, numerically validated view extrapolation, mesh reconstruction, generalization across different scenes and correlation between scene's variability and reconstruction quality (**NeRF, Python, PyTorch**).
- Researched and implemented a font classifier via semi-supervised learning used for document verification (**PyTorch**).

Photomath (Google) – Software Engineer Intern — July 2022 – Oct 2022
- Developed and deployed cloud services that parse, transform, enrich, and deliver events from Photomath's textbook solutions database (**Python, GCP, Datastream, Pub/Sub, Dataflow, Cloud Run, GitHub Actions, Spark**).

Memgraph – Software Engineer — July 2020 – Oct 2021
- Designed a **PostgreSQL** database schema, wrote user stories, and feature specifications for Memgraph Cloud platform.
- Implemented Memgraph Cloud's backend that manages **AWS EC2** instances, and supports monthly user billing based on user usage (**node.js, TypeScript, Express, Sequelize, PostgreSQL**).
- Set up **Elastic Stack** on **AWS EC2** to analyze application logs via **AWS CloudWatch** in **Kibana** dashboards.
- Implemented real-time graph data visualization for geospatial data on top of OpenStreetMap (**TypeScript, Leaflet**).

Skills

Languages:	**Python, TypeScript, bash**
Technologies:	**git, Linux, PyTorch, OpenCV, numpy, Docker, LaTeX, Django, GCP, AWS**
Other:	**Machine Learning, Computer Vision, Data Structures, Distributed Systems, Product Strategy**

Extracurricular

Machine Learning competition – Musical Instrument Classification — 2023
- **Led a team of four. Won the 2nd place and 1st place in model performance.**
- Trained a model that labels 11 musical instruments from an input audio signal, developed in PyTorch using deep learning, digital signal processing, audio feature engineering, and computer vision style spectrogram/image representations derived from raw audio. Deployed the trained model behind a FastAPI inference service.
- LUMEN Data Science is the largest machine learning competition in the Croatia.

A Squeeze of LIME, a Pinch of SHAP – an NLP paper — 2023
- Wrote and published an NLP paper that analyzes the explainability of the trained model with LIME and SHAP, model-agnostic and post-hoc methods with two other authors.
- Trained a DeBERTa-based model for classification of sentence coherence.

Machine Learning competition – Computer Vision GeoGuessr — 2022
- **Led a finalist team of three. Won the 2nd place in model performance.**
- Trained a model that predicts the lat/lng location of Google Street View images, developed in PyTorch using deep learning, computer vision, and geospatial feature engineering. Deployed the trained model behind a FastAPI inference service.
- Wrote both competition report and technical documentation.
- LUMEN Data Science is the largest machine learning competition in the Croatia.

Paper implementation – driver fatigue detection in an EEG-based system — 2022
- Reproduced paper's baseline and methods. Trained the model that matches the reported performance.
- Analyzed and extracted entropy features from driver's EEG data.
- Trained and compared four different models, achieving +1% higher performance than reported in the paper.

Other

Certificate in Advanced Rhetoric and Public Speaking (Heureka) — 2026
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Attended Croatian Machine Learning Workshop (CMLW) — 2024
Certificate in Soft Skills (EESTEC) — 2019